Digital SAT Math

Topic 2: Exponential Functions - Growth and Decay

Practice Questions

Total: 60 Questions

Format: Digital SAT Style — MCQ & Grid-In

Difficulty: Easy (Q1-20) • Medium (Q21-45) • Hard (Q46-60)

Q1.

DIFFICULTY: Easy

$f(x) = 3(2)^x$. What is f(4)?

A) 24
B) 30
C) 48
D) 96

Q2.

DIFFICULTY: Easy

A population of bacteria doubles every hour. If there are initially 500 bacteria, which expression gives the number after t hours?

A) 500 + 2t
B) $500(2)^t$
C) 500 * 2t
D) $2(500)^t$

Q3.

DIFFICULTY: Easy

$g(x) = 4(0.5)^x$. Which of the following describes the behavior of g as x increases?

A) Increasing, because the base is greater than 1
B) Decreasing, because the base is between 0 and 1
C) Decreasing, because the coefficient is positive
D) Increasing, because the coefficient is 4

Q4.

DIFFICULTY: Easy

The value V, in dollars, of a car t years after purchase is given by $V = 18{,}000(0.85)^t$. What is the initial value of the car?

A) $0.85
B) $15,300
C) $18,000
D) $21,176

Q5.

DIFFICULTY: Easy

Which of the following represents exponential growth?

A) $f(x) = 3x + 5$
B) $f(x) = x^3$
C) $f(x) = 3(1.2)^x$
D) $f(x) = 3(0.8)^x$

Q6.

DIFFICULTY: Easy

$h(t) = 100(0.9)^t$. What is h(0)?

A) 0
B) 0.9
C) 90
D) 100

Q7.

DIFFICULTY: Easy

$f(x) = 5(3)^x$. What is f(0) + f(1)?

A) 8
B) 10
C) 15
D) 20

Q8.

DIFFICULTY: Easy

An account earns 4% interest per year, compounded annually. Starting with $1,000, which gives the account value A(t) after t years?

A) $A(t) = 1{,}000 + 40t$
B) $A(t) = 1{,}000(1.04)^t$
C) $A(t) = 1{,}000(0.04)^t$
D) $A(t) = 1{,}000(4)^t$

Q9.

DIFFICULTY: Easy

A radioactive substance has a half-life of 10 years. A sample starts with 80 grams. Which expression gives the remaining amount after t years?

A) $80 - 4t$
B) $80(0.5)^{t/10}$
C) $80(2)^t$
D) $80(10)^t$

Q10.

DIFFICULTY: Easy

$f(x) = a(b)^x$ passes through (0, 6) and (1, 18). What is b?

A) 2
B) 3
C) 6
D) 12

Q11.

DIFFICULTY: Easy

A laptop costs $800 new and decreases in value by 20% each year. Which expression gives its value after 3 years?

A) 800 - 3(0.20)(800)
B) $800(0.8)^3$
C) $800(1.2)^3$
D) $800(0.2)^3$

Q12.

DIFFICULTY: Easy

$p(t) = 200(1.05)^t$. Which best describes p?

A) A linear function with slope 200
B) An exponential function with 5% growth per period
C) An exponential function with 5% decay per period
D) A quadratic function with vertex at (0, 200)

Q13.

DIFFICULTY: Easy

What is the y-intercept of the graph of $f(x) = 7(4)^x$?

A) (0, 1)
B) (0, 4)
C) (0, 7)
D) (0, 28)

Q14.

DIFFICULTY: Easy

A city's population of 50,000 grows by 3% each year. Which gives the population after 1 year?

A) 50,000 + 0.03
B) 50,000(0.03)
C) 50,000(1.03)
D) 50,000(3)

Q15.

DIFFICULTY: Easy

$f(x) = 2^x$ and $g(x) = 2^{x+3}$. Which of the following is true?

A) g(x) = f(x) + 3
B) g(x) = 8 * f(x)
C) g(x) = f(x) - 3
D) g(x) = f(x) / 8

Q16.

DIFFICULTY: Easy

Which best explains why the table below could represent an exponential function?

x | y

0 | 1

1 | 3

2 | 9

3 | 27

A) Each y-value is 3 times the previous y-value
B) Each y-value increases by 2
C) The differences between consecutive y-values are constant
D) The y-intercept is 0

Q17.

DIFFICULTY: Easy

$f(x) = 5(2)^x$. What is f(3)?

[GRID-IN — Enter your answer below]

Q18.

DIFFICULTY: Easy

$d(t) = 1{,}200(0.75)^t$ gives the amount of medicine, in mg, remaining t hours after administration. By what percent does the medicine decrease each hour?

A) 25%
B) 75%
C) 80%
D) 125%

Q19.

DIFFICULTY: Easy

$f(x) = 6(b)^x$, where b is a positive constant. If f(2) = 54, what is b?

A) 3
B) 6
C) 9
D) 27

Q20.

DIFFICULTY: Easy

A colony of ants triples every 2 weeks. Starting with 400 ants, how many are in the colony after 6 weeks?

A) 1,200
B) 3,600
C) 10,800
D) 32,400

Q21.

DIFFICULTY: Medium

$f(x) = a(b)^x$ with f(0) = 12 and f(3) = 96. What is b?

A) 2
B) 3
C) 4
D) 8

Q22.

DIFFICULTY: Medium

The table below shows values of a function. Which equation best models the data?

x | y

0 | 5

1 | 10

2 | 20

3 | 40

A) y = 5x + 5
B) $y = 5(2)^x$
C) $y = 2(5)^x$
D) y = 10x

Q23.

DIFFICULTY: Medium

$f(t) = 3,000(1 + r)^t$ models a bank account. If the amount after 2 years is $3,363, what is the approximate value of r?

A) 0.03
B) 0.06
C) 0.10
D) 0.08

Q24.

DIFFICULTY: Medium

A substance decays so that 1/8 of it remains after 30 minutes. For $Q(t) = Q\_0 * b^t$ (t in minutes), what is b?

A) $(1/2)^{1/10}$
B) $(1/8)^{1/30}$
C) $(1/2)^{1/30}$
D) 1/8

Q25.

DIFFICULTY: Medium

In $f(x) = a * b^x$, the value of f increases by 40% for each 1-unit increase in x. Which could be the value of b?

A) 0.4
B) 0.6
C) 1.04
D) 1.4

Q26.

DIFFICULTY: Medium

$f(x) = 500(1.06)^x$ and $g(x) = 500(1.06)^{x-3}$. Which is true?

A) g(x) = f(x) - 3
B) g(x) = f(x)/f(3)
C) g(x) = f(x - 3)
D) g(x) = f(x) * f(3)

Q27.

DIFFICULTY: Medium

$N(t) = 2{,}500(1.08)^t$ models streaming subscribers t months after launch. Approximately how many subscribers after 12 months?

A) 2,500
B) 3,000
C) 5,000
D) 6,326

Q28.

DIFFICULTY: Medium

$f(x) = 4^x$. Which is equivalent to $f(x + 2)$?

A) f(x) + 16
B) 16 * f(x)
C) f(x) + 2
D) 2 * f(x)

Q29.

DIFFICULTY: Medium

A 640 mg sample loses half its mass every 4 days. Which gives the mass m, in mg, remaining after d days?

A) $m = 640(0.5)^d$
B) $m = 640(0.5)^{d/4}$
C) $m = 640(0.5)^{4d}$
D) m = 640 - 80d

Q30.

DIFFICULTY: Medium

$f(x) = a * b^x$ satisfies f(1) = 10 and f(4) = 80. What is a?

A) 2
B) 4
C) 5
D) 8

Q31.

DIFFICULTY: Medium

$y = 2^x$ is transformed to produce $y = 2^x - 8$. Which describes this transformation?

A) Shift right 8 units
B) Shift left 8 units
C) Shift down 8 units
D) Vertical stretch by factor 8

Q32.

DIFFICULTY: Medium

$f(x) = 200(1.04)^x$ and $g(x) = 200(1.04)^x + 100$. Which is true?

A) g has the same growth rate as f, but a higher initial value.
B) g grows at a rate 4% higher than f.
C) g has the same initial value as f, but decays.
D) g is not an exponential function.

Q33.

DIFFICULTY: Medium

$P(t) = 12{,}000(1.025)^t$ models a town's population t years after 2010. In what year will the population first exceed 15,000?

A) 2019
B) 2020
C) 2021
D) 2022

Q34.

DIFFICULTY: Medium

Which of the following does NOT represent exponential decay?

A) $y = 5(0.3)^x$
B) $y = (3/4)^x$
C) $y = 2(1.1)^x$
D) $y = 100(0.99)^x$

Q35.

DIFFICULTY: Medium

In $P(t) = P_0 * e^{kt}$, if $P_0 = 100$ and $P(5) = 200$, what is the approximate value of k?

A) 0.07
B) 0.14
C) 0.20
D) 0.40

Q36.

DIFFICULTY: Medium

$f(x) = 3^x$ and $g(x) = 3^{2x}$. Which is equivalent to g(x)?

A) $2 * f(x)$
B) $[f(x)]^2$
C) $f(x) + f(x)$
D) $f(x + 2)$

Q37.

DIFFICULTY: Medium

A car bought for $24,000 decreases in value by 15% each year. After how many complete years will its value first fall below $12,000?

A) 4
B) 5
C) 6
D) 7

Q38.

DIFFICULTY: Medium

$f(x) = 8(2)^{x/3}$ can be written as $f(x) = a * b^x$. What is b?

A) 2
B) $2^{1/3}$
C) 2^3
D) 8

Q39.

DIFFICULTY: Medium

$5,000 is invested at 6% annual interest compounded annually. How much more is in the account after 10 years than after 5 years? (Round to nearest dollar.)

A) $1,767
B) $2,263
C) $2,552
D) $3,439

Q40.

DIFFICULTY: Medium

The table shows values of an exponential function f.

x | f(x)

0 | a

1 | 12

2 | 36

What is a?

[GRID-IN — Enter your answer below]

Q41.

DIFFICULTY: Medium

A yeast culture triples every hour. Starting with n cells, which expressions give the count after t hours?

I. $n * 3^t$

II. $n * (1 + 2)^t$

A) I only
B) II only
C) I and II
D) Neither

Q42.

DIFFICULTY: Medium

$f(x) = 2(3)^x + 5$. What is the equation of the horizontal asymptote of y = f(x)?

A) y = 0
B) y = 2
C) y = 5
D) y = 7

Q43.

DIFFICULTY: Medium

$W(t) = 4{,}000(0.92)^t$ gives gallons of water remaining in a tank at time t minutes. By approximately what percentage is the water level reduced in the first 10 minutes?

A) 8%
B) 43%
C) 57%
D) 80%

Q44.

DIFFICULTY: Medium

Both $y = 3^x$ and $y = 3^{-x}$ are graphed in the xy-plane. Which describes the relationship?

A) $y = 3^{-x}$ is $y = 3^x$ reflected over the x-axis.
B) $y = 3^{-x}$ is $y = 3^x$ reflected over the y-axis.
C) $y = 3^{-x}$ is $y = 3^x$ shifted left.
D) The graphs are identical.

Q45.

DIFFICULTY: Medium

$A(t) = 500(0.5)^{t/6}$ gives drug amount in mg remaining t hours after a 500 mg dose. After how many hours does approximately 62.5 mg remain?

A) 12
B) 18
C) 24
D) 30

Q46.

DIFFICULTY: Hard

$f(x) = 20(1.3)^{x+k}$ and $g(x) = 20(1.3)^x$, where k is a positive constant. Which is equivalent to f(x) in terms of g(x)?

A) $f(x) = g(x) + k$
B) $f(x) = (1.3)^k * g(x)$
C) $f(x) = k * g(x)$
D) $f(x) = g(x + k)$

Q47.

DIFFICULTY: Hard

$f(x) = a * b^x$ with f(2) = 36 and f(5) = 972. What is a?

A) 3
B) 4
C) 9
D) 12

Q48.

DIFFICULTY: Hard

$N(t) = 800 / (1 + 3e^{-0.4t})$ models how many people have heard a rumor after t days. Approximately how many after 5 days?

A) 267
B) 400
C) 491
D) 569

Q49.

DIFFICULTY: Hard

$f(x) = 16(2)^{x/4}$. Which equivalent form displays the value of x at which f(x) = 32 as a constant or coefficient?

A) $f(x) = 16 * 2^{x/4}$
B) $f(x) = 32 * 2^{(x-4)/4}$
C) $f(x) = 2^{(x+16)/4}$
D) $f(x) = 16^{x/4}$

Q50.

DIFFICULTY: Hard

$p(t) = 3{,}200(1 + r)^{12t}$ where r is the monthly rate and t is years. If the annual rate is 12%, what is r?

A) 0.01
B) 0.12
C) 1.00
D) 1.12

Q51.

DIFFICULTY: Hard

$f(x) = 10(1.5)^{x+2}$ and $g(x) = 10(1.5)^x * c$ define the same function. What is c?

[GRID-IN — Enter your answer below]

Q52.

DIFFICULTY: Hard

$g(x) = 5 * 4^x - 60$. How many times does the graph of $y = g(x)$ cross the x-axis?

A) 0
B) 1
C) 2
D) 3

Q53.

DIFFICULTY: Hard

$f(x) = 7(1.06)^{x/12}$ can be written as $f(x) = 7b^x$. What is the approximate value of b?

A) 1.005
B) 1.06
C) 1.072
D) 1.12

Q54.

DIFFICULTY: Hard

$A(t) = 400(1.2)^t$ and $B(t) = 1{,}600(1.05)^t$ model two insect populations (t in years). After approximately how many years does Population A first exceed Population B?

A) Between 10 and 11 years
B) Between 15 and 16 years
C) Between 20 and 21 years
D) Between 25 and 26 years

Q55.

DIFFICULTY: Hard

$f(t) = a * b^t$ passes through (0, 8) and (3, 1). What is b?

A) 1/2
B) 1/3
C) 1/4
D) 1/8

Q56.

DIFFICULTY: Hard

$N(t) = 100(2)^{t/3}$ and $M(t) = 400(2)^{t/6}$ model two bacterial colonies (t in hours). At what time t are they equal in size?

[GRID-IN — Enter your answer below]

Q57.

DIFFICULTY: Hard

$f(x) = a * 3^{bx}$ has y-intercept 5 and passes through (2, 45). What is b?

A) 1/2
B) 1
C) 3/2
D) 2

Q58.

DIFFICULTY: Hard

The graph of $f(x) = c * d^x$ is decreasing and passes through (0, 4) and (2, 1). Which must be true?

A) $c < 1$ and $0 < d < 1$
B) $c > 1$ and $0 < d < 1$
C) $c > 1$ and $d > 1$
D) $c < 0$ and $d > 1$

Q59.

DIFFICULTY: Hard

$f(x) = a(b)^x + d$ has horizontal asymptote $y = 3$ and passes through (0, 7) and (1, 11). What is b?

A) 2
B) 3
C) 4
D) 8

Q60.

DIFFICULTY: Hard

An exponential function f satisfies $f(n+1)/f(n) = 1.5$ for all positive integers n. If $f(1) = 6$, what is $f(5)$?

[GRID-IN — Enter your answer below]